

COMPARATIVE ADVANTAGE PROBLEM SET

ECON101—INTRODUCTION TO MICROECONOMICS.

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Problem 1

- (a) Rakhim can wax a car in 20 minutes or wash a car in 60 minutes. Sultan can wax a car in 15 minutes or wash a car in 30 minutes. What is each person's opportunity cost of washing a car? Who has a comparative advantage in washing cars?
- (b) Rakhim can wax 4 cars per day or wash 12 cars. Sultan can wax 3 cars per day or wash 6 cars. What is each person's opportunity cost of washing a car? Who has a comparative advantage in washing cars?

Problem 2

Akmaral and Abuzer are auto mechanics. Akmaral takes 4 hours to replace a clutch and 2 hours to replace a set of brakes. Abuzer takes 6 hours to replace a clutch and 2 hours to replace a set of brakes. State whether anyone has an absolute advantage at either task and, for each task, identify who has a comparative advantage.

Problem 3

Consider a society consisting only of Daniyar, who allocates his time between sewing dresses and baking bread. Each hour he devotes to sewing dresses yields 4 dresses and each hour he devotes to baking bread yields 8 loaves of bread.

- (a) If Daniyar works a total of 8 hours per day, graph her production possibilities frontier.
- (b) Using your graph, which of the points listed below are attainable and/or efficient?
- 28 dresses per day, 16 loaves per day.
 - 16 dresses per day, 32 loaves per day.
 - 18 dresses per day, 24 loaves per day.

Problem 4

Suppose that in the previous problem a new sewing machine is introduced that enables Daniyar to sew 8 dresses per hour rather than only 4.

- (a) Show how this development shifts her production possibilities curve.
- (b) Indicate if the following points are attainable and/or efficient before and after the introduction of the sewing machine.

- 16 dresses per day, 48 loaves per day.
 - 24 dresses per day, 16 loaves per day.
- (c) Explain what is meant by the following statement: “An increase in productivity with respect to any one good increases our options for producing and consuming all other goods.”

Problem 5

Medina can pick 4 kilograms of coffee beans in an hour or gather 2 kilograms of nuts. Nuray can pick 2 kilograms of coffee beans in an hour or gather 4 kilograms of nuts. Each works 6 hours per day.

- (a) Together, what is the maximum number of kilograms of coffee beans the two can pick in a day? What is the maximum number of kilograms of nuts the two can gather in a day?
- (b) Now suppose Medina and Nuray were gathering the maximum number of kilograms of nuts when they decided that they would like to begin picking 8 kilograms of coffee per day. Who would pick the coffee, and how many kilograms of nuts would they still be able to gather?
- (c) Would it be possible for Medina and Nuray in total to gather 26 kilograms of nuts and pick 20 kilograms of coffee each day? If so, how much of each good should each person pick?
- (d) Is the point at 30 kilograms of coffee per day, 12 kilograms of nuts per day an attainable point? Is it an efficient point?
- (e) On a graph with kilograms of coffee per day on the vertical axis and kilograms of nuts per day on the horizontal axis, show all the points you identified in parts a–d.

Problem 6

Refer to the two-person economy described in problem 5.

- (a) Suppose that Medina and Nuray could buy or sell coffee and nuts in the world market at a price of \$2 per kilogram for coffee and \$2 per kilogram for nuts. If each person specialized completely in the good for which each had a comparative advantage, how much could they earn by selling all they produce?
- (b) At the prices just described, what is the maximum amount of coffee Medina and Nuray could buy in the world market with the income they earned? What is the maximum amount of nuts? Would it be possible for them to consume 40 kilograms of nuts and 8 kilograms of coffee each day?
- (c) In light of their ability to buy and sell in world markets at the stated prices, show on the same graph all combinations of the two goods it would be possible for them to consume.
- (d) *More difficult.* Define the *terms of trade* as the *relative price* of coffee per nuts in the world market, i.e., the ratio between the price of coffee and the price of nuts. In the previous parts a–c, this price ratio was equal to 1. At what interval of price ratios would it still be beneficial for Medina and Nuray to specialize in the goods for which each has a comparative advantage?

Problem 7

Which of the following jobs is likely to be the most vulnerable to outsourcing: and in-home health care provider, a hairstylist, or a computer programmer? Explain.

Problem 8

Consider an economy with only two goods, x and y . Suppose that the production of both x and y depends only on labor input and that the production functions for these goods are:

$$x = \sqrt{l_x},$$
$$y = \sqrt{l_y},$$

where l_x and l_y are the amounts of labor devoted to the production of x and y respectively. Suppose that total labor supply, i.e., the total amount of working hours, is limited by

$$l_x + l_y = 100.$$

- (a) Derive the production possibilities frontier for this economy.
- (b) How does the opportunity cost of good x change along the production possibilities frontier? Give an intuitive explanation first, and then try to give, to the best of your ability and knowledge, a formal mathematical explanation.